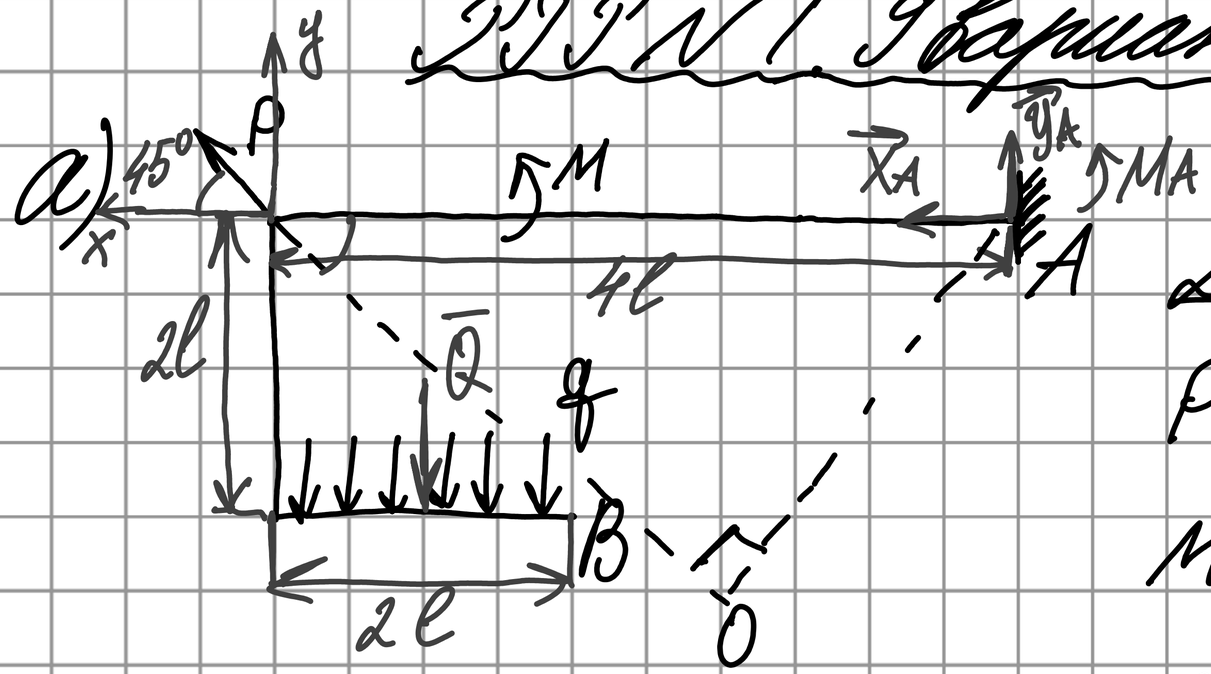
РГР N 1 Вариант

Мовсесян Игорь
R3280



Дано:

$P = 4\,кН$

$M = 10\,кН\cdot м$

$q = 1\,кН/м$

$\ell = 1\,м$

Решение:

$Q = 2q\ell$

Запишем систему:

	$\vec{X}_A$	$\vec{y}_A$	$\vec{P}$	$\vec{Q}$	M	M_A	
x	X_A	0	$P\cos 45°$	0	0	0	$= 0$
y	0	y_A	$P\sin 45°$	Q	0	0	$= 0$
M	0	0	$4\ell P\sin 45°$ (P·OA)	$Q\cdot 3\ell$	M	M_A	$= 0$

$\approx 2{,}83\,кН$

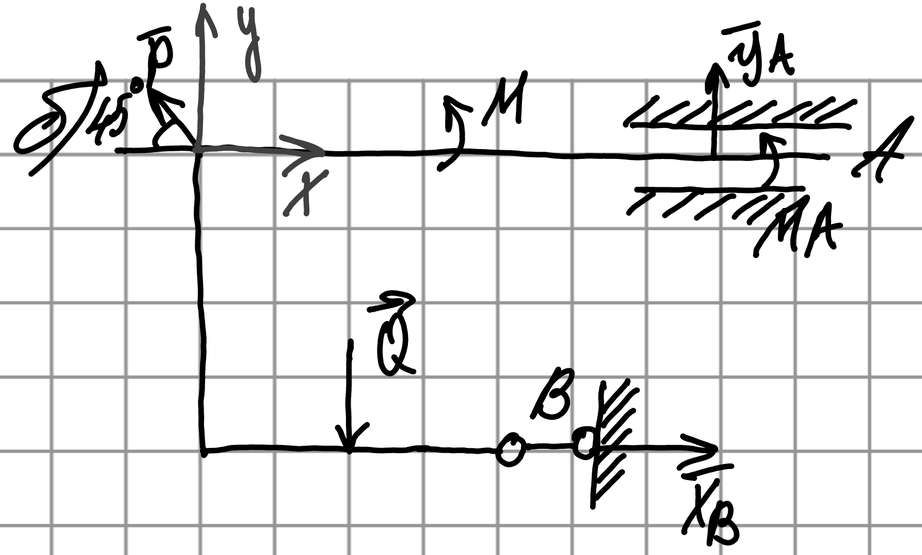
$X_A = -P\cos 45° = -2\sqrt{2}\,кН$

$y_A = -P\sin 45° = -2\sqrt{2}\,кН \approx -2{,}83\,кН$

$M_A = -M - Q\cdot 3\ell - 4\ell P\sin 45° =$

$= -M - 6q\ell^2 - 4\ell P\sin 45° = -16 - 16\sqrt{2} =$

$= -16(1+\sqrt{2}) \approx -27{,}31\,кН\cdot м$



	X_B	P	y_A	Q	M	M_A	
X	X_B	$-P\cos 45°$	0	0	0	0	$=0$
y	0	$P\sin 45°$	y_A	$-Q$	0	0	$=0$
M	$2\ell X_B$	$4\ell P\sin 45°$	0	$3\ell Q$	M	M_A	$=0$

$$X_B = P\cos 45° = 2\sqrt{2}\ кН \approx 2{,}83\ кН$$

$$y_A = Q - P\sin 45° = 2q\ell - P\sin 45° = 2 - 2\sqrt{2}\ кН = 2(1-\sqrt{2})\ кН \approx 0{,}83\ кН$$

$$M_A = -M - 3\ell Q - 2\ell X_B - 4\ell P\sin 45° \ (=)$$

$$(=) -M - 6\ell q^2 - 2\ell X_B - 2\sqrt{2}\ell P \approx 32{,}97\ кН\cdot м$$